



Bridging Radiology and Pathology

A DICOM-based Framework for Multimodal Mapping and Integrated Visualization

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Abstract. Accurate disease diagnosis depends on effective collaboration between medical specialties, yet departments often use distinct data systems and proprietary formats. This heterogeneity hinders joint analysis and integration of complementary diagnostic information. The use of separate viewers for each modality further restricts cross-specialty collaboration. Although multimodal integration, particularly between radiology and pathology, has demonstrated potential for identifying novel biomarkers, it still relies heavily on manual, time-consuming data pairing. This project introduces an interdisciplinary toolbox that can operate within the Kaapana framework or as a standalone tool to bridge radiology and pathology. By linking modality-specific viewers and extending them with automated image registration and alignment, the platform enables efficient, scalable multimodal analysis. The integrated environment promotes reproducible workflows, accelerates cross-disciplinary research, and facilitates deeper insights into disease mechanisms and patient care. The method will be integrated in the Kaapana toolkit and as standalone package <https://github.com/MIC-DKFZ/combinedmodalityviewer>.

1 Introduction

The increasing availability of multimodal medical data, particularly the combination of radiological and pathological data, offers great potential for comprehensive disease characterization. However, integrating and visualizing these data streams is

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technically challenging due to fundamental differences in their scale and acquisition. Currently, few systems support the spatially registered and interactive inspection of these distinct modalities within a unified environment. Radiology has undergone extensive digitization, supported by standardized DICOM-based infrastructures and well-established analytical tools [1]. In contrast, digital pathology is still transitioning toward routine digital adoption, and specialized microscopy viewers often operate in isolation from radiological systems [2]. This data fragmentation limits consistent multimodal exploration and hinders the interpretability of complex algorithmic pipelines. In response, we developed a lightweight, modular viewer for the synchronized visualization of spatially registered radiological and histopathological data. This system integrates established open-source components into a single, coherent web interface, demonstrating a practical and extensible solution for multimodal data exploration. The proposed viewer serves as a bridge between radiological and pathological workflows and establishes a foundation for future research.

2 Methods

For improving multimodal image evaluation, two requirements must be fulfilled: the modalities must be mapped onto each other, and the linked dataset must be visualized appropriately. Both branches are tackled with our proposed framework. This section presents the methods that are used to link radiology and pathology images onto each other, as well as the visualization tools to visualize the linked datasets.

2.1 Data preprocessing

The integration of both modalities requires preprocessing, including DICOM conversion of whole slide images (WSIs) and definition of their anatomical origin.

2.1.1 Dicom conversion. A unified data format is essential for integrating multiple imaging modalities. In radiology, DICOM is the established standard, and with the DICOM-WSI extension it now also supports pathology [3]. Our pipeline therefore requires both modalities in DICOM format and linked through the corresponding DICOM tags². To achieve this, we employ the `wsicomizer`³, which is also natively integrated into other multimodal imaging platforms such as Kaapana [4]. The tool can be used within Kaapana or independently, supported by the comprehensive documentation.

2.1.2 Localization of WSI in radiology. Once both modalities are available in the picture archiving and communication system (PACS), the mapping pipeline can be initiated after verifying data completeness. A key step is aligning WSIs with their corresponding radiological images by resolving the anatomical region. The routine

²<https://dicom.nema.org/medical/dicom/current/output/html/part18.html>

³<https://github.com/imi-bigpicture/wsidicomizer>